Nonlinear Model Predictive Control for Quadrotors Using Simultaneous Optimization Approach

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***Abstract*—The quadrotor has found extensive applications across a spectrum of fields, including environmental research, military operations, transportation, and firefighting. The design of its control system can be approached using various methodologies such as sliding mode control, backstepping, and model predictive control (MPC). The primary objective of the control system is to achieve trajectory tracking while simultaneously avoiding collisions with obstacles. In this paper, a nonlinear model predictive control (NMPC) strategy is adopted for calculating control inputs that guide the quadrotor along a desired trajectory while ensuring obstacle avoidance. The obstacle avoidance constraints that are formulated pose a nonlinear and intricate challenge, rendering the optimization problem complex for the nonlinear program solver to address. To tackle this, linear constraints representing hyperplanes between the quadrotor and obstacles are employed. These hyperplanes dynamically adjust as the quadrotor moves, thereby averting collisions effectively. The application of NMPC to quadrotor control, as demonstrated in this paper, underscores the suitability and practicality of the proposed approach. It successfully showcases the efficacy of NMPC in facilitating quadrotor control while navigating through complex environments.**

Keywords—component, formatting, style, styling, insert (*key words*)

I. Introduction

In recent years, unmanned aerial vehicles (UAVs) have found widespread applications across various domains, including military and commercial sectors for surveillance, exploration, aerial inspection, photography, and water quality management. These UAVs can be broadly categorized into two main types: fixed-wing and rotary-wing UAVs [1-2]. Among the rotary-wing UAVs, the quad-copter or quadrotor, equipped with four rotors, has gained popularity for small payload applications and is utilized in numerous fields. Furthermore, UAV technologies have advanced to incorporate specialized vision and intelligence methods, encompassing functions such as object detection, path planning, and object tracking [1]. These intelligent capabilities have prompted the development of a methodology that empowers UAVs to autonomously generate collision-free waypoints in their immediate vicinity.

Various approaches have been developed to tackle control design for quadrotors. The PID control approach was employed in [3-4], while an LQR controller was applied in [4]. These controllers are grounded in linear control methodologies. However, in practical scenarios, UAV models exhibit nonlinearity, prompting the use of alternative approaches such as backstepping control [5], nonlinear H_∞ control [6], and sliding mode control (SMC) [5].

Recently, model predictive control (MPC) has been harnessed to manage UAVs, primarily due to its adeptness in compensating for highly nonlinear model systems and accommodating various constraints. Viewing UAVs as agents, a distributed guidance and control algorithm for reconfiguring swarms comprising hundreds to thousands of agents with limited communication and computational capabilities was proposed in [2,7]. This algorithm addresses optimal assignment and collision-free trajectory generation. To achieve collision-free trajectories, hyperplane movement is accomplished. Additionally, the optimization problem is addressed using sequential convex programming [7]. In [8], the optimization problem for optimal trajectory flight is formulated, incorporating collision exclusion areas for obstacle avoidance constraints. The optimization problem is tackled through the utilization of the multiple shooting discretization approach combined with the Ipopt solver [9].

In this paper, we proposed to combine the simultaneous optimization approach and the hyperplane moving for collision free method in the framework computation of the nonlinear model predictive control (NMPC) for guidance of the quadrotor to the desired paths for monitoring water quality. It is due to the fact that the hyperplane moving method utilizes the linear constraints for obstacle avoidance, the resulting NLP is easier to be addressed by standard NLP solvers enabling the efficacy of NMPC implementation.

II. Dynamic Model of Quadrotor and Optimal Control Problem

A. Dynamic model of Quadrotor and optimal control problem

The quadrotor frames are shown in Fig.1. We define R as a transformation matrix between the body-fixed frame of the UAV and the inertial one; ϕ, θ, ψ are the angles of roll, pitch, and yaw, respectively. The relationships between the yaw, pitch, and roll angles in both frames are established through the matrix R as follows [1-2]

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$$R = \begin{pmatrix} C\psi C\theta & C\psi S\theta S\phi - S\psi C\phi & C\psi S\theta C\phi + S\psi S\phi \\ S\psi C\theta & S\psi S\theta S\phi + C\psi C\phi & S\psi S\theta C\phi - C\psi S\phi \\ -S\theta & C\theta S\phi & C\theta C\phi \end{pmatrix} \quad (1)$$

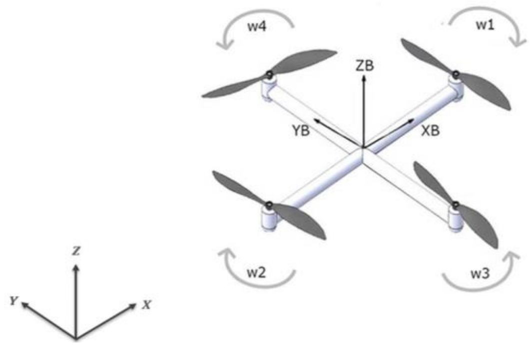

Fig. 1. The body and inertial frames of the quadrotor

In the matrix, the symbol of C and S denote cosine and sine function, respectively. Similarly, the transformation between Euler angles rates in inertial frame, $\dot{\eta} = [\dot{\phi}, \dot{\theta}, \dot{\psi}]$, and the angular velocities measured in body frames, $\Omega = [p, q, r]$, is

$$\Omega = W_\eta \dot{\eta} \quad (2)$$

with

$$W_\eta = \begin{pmatrix} 1 & 0 & -\sin\,\theta \\ 0 & \cos\,\phi & \sin\,\phi\cos\,\theta \\ 0 & -\sin\,\phi & \cos\,\phi\cos\,\theta \end{pmatrix} \quad (3)$$

Since the roll and pitch angles are small, W_η can be approximated to the identity Matrix $I_{3\times3}$. For this, we have

$$\Omega = \dot{\eta} = [\dot{\phi}, \dot{\theta}, \dot{\psi}] \quad (4)$$

Now, we employ the Newton-Euler method to derive the dynamic model of the quadrotor. The acceleration of the quadrotor is the result of the total thrust of rotors T_B and the gravitational force G.

$$m\ddot{\zeta} = G + RT_B$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = -g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + R\frac{T_B}{m} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (5)$$

where $\zeta = [x \quad y \quad z]^T$ is the position of the quadrotor in the inertial frame.

The total thrust vector (T_B) applied in the body axes can be written by combining the forces of four rotors [2]

$$T_B = \sum_{i=1}^{4} |F_i| = b \sum_{i=1}^{4} \omega_i^2 \quad (6)$$

where b is the thrust constant.

From equation (5), the model equations relating the position of UAV with the torque and angles of ϕ, θ, ψ are

$$\ddot{x} = (\cos\,\phi \cdot \sin\,\theta \cdot \cos\,\psi + \sin\,\phi \cdot \sin\,\psi) \cdot \frac{T_B}{m}$$

$$\ddot{y} = (\cos\,\phi \cdot \sin\,\theta \cdot \sin\,\psi - \sin\,\phi \cdot \cos\,\psi) \cdot \frac{T_B}{m} \quad (7)$$

$$\ddot{z} = -g + (\cos\,\phi \cdot \cos\,\theta) \cdot \frac{T_B}{m}$$

Furthermore, within the body frame, the equation governing rotational motion is deduced from the equilibrium between the external torque exerted on the quadrotor, denoted as τ_B, and the combined effect of the angular acceleration of the inertia $I\dot{\Omega}$ and the centripetal forces represented by the cross product $\Omega \wedge I\Omega$

$$\tau_B = I\Omega + \Omega \wedge I\Omega \quad (8)$$

With $I = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}$

Additionally, we introduce τ_ϕ, τ_θ, and τ_ψ as the roll torque, pitch torque, and yaw torque, respectively. These torques are linked to the angular velocity of the four rotors through the following equation [1-2]

$$\tau_B = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} lb(-\omega_2^2 + \omega_4^2) \\ lb(-\omega_1^2 + \omega_3^2) \\ d(-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^3) \end{bmatrix}$$

where l is the distance between the rotor and the center of mass of the quadrotor; b is the thrust rotor factor while d stands for the drag factor. The control vector is now defined as

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} T_B \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} b & b & b & b \\ 0 & -lb & 0 & lb \\ -lb & 0 & lb & 0 \\ -d & d & -d & d \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \quad (9)$$

Replace it into equation (8), we have

$$\begin{aligned} \ddot{\phi} &= \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) + \frac{u_2}{I_x} \\ \ddot{\theta} &= \dot{\phi}\dot{\psi}\left(\frac{I_z - I_x}{I_y}\right) + \frac{u_3}{I_y} \\ \ddot{\psi} &= \dot{\theta}\dot{\phi}\left(\frac{I_x - I_y}{I_z}\right) + \frac{u_4}{I_z} \end{aligned} \quad (10)$$

The primary objective of optimal control is to navigate the quadrotor along the desired trajectory while simultaneously ensuring avoidance of collisions with obstacles. To achieve this goal, the formulation of the optimal control problem (OCP) is expressed as follows:

$$\min_{u(t),z(t)} \varphi\big(\boldsymbol{z}(t), \boldsymbol{u}(t)\big)$$

$$\frac{dz(t)}{dt} = \boldsymbol{F}\big(\boldsymbol{z}(t), \boldsymbol{u}(t)\big); \boldsymbol{z}(t_0) = \boldsymbol{z}_0 \qquad (11)$$

$$\boldsymbol{G}\big(\boldsymbol{z}(t), \boldsymbol{u}(t)\big) \le 0$$

$$\boldsymbol{z}(t)_{\boldsymbol{min}} \le \boldsymbol{z}(t) \le \boldsymbol{z}(t)_{\boldsymbol{max}}$$

$$\boldsymbol{u}(t)_{\boldsymbol{min}} \le \boldsymbol{u}(t) \le \boldsymbol{u}(t)_{\boldsymbol{max}}$$

where $\boldsymbol{z}(t)$ represent as a vector of state variables in (10) while $\boldsymbol{u}(t)$ is a vector of control variables; $\boldsymbol{F}$ represents equations in (7) and (10).

B. *Obstacle avoidance hyperplane moving method*

In this paper, obstacle avoidance constraints are applied to ensure that collision is free with the obstacles and / or other quadrotor. To do this, for each quadrotor, the following constraint is applied.

$$\left\|p_j(t) - p_i(t)\right\|_2 \ge R_{col}$$

where $p_j(t) = [x_j(t) \quad y_j(t) \quad z_j(t)]^T$ is the position of the quadrotor k and p_i is the position of obstacles or other quadrotors; R_{col} is the diameter of the ball centering the obstacles or quadrotors. It is well known that the above constraints are nonlinear and non convex, rendering the OCP complicated to be solved by standard OCP solvers. To deal with this issue, these constraints are linearized around their current positions (i.e., $\bar{p}_j$ and $\bar{p}_i$). The linearized constraints are [7]

$$\big(\bar{p}_j(t) - \bar{p}_i(t)\big)^T \left(p_j(t) - \bar{p}_{i,k}\right) \ge R_{col}\left\|\bar{p}_j(t) - \bar{p}_i(t)\right\|_2 \quad (12)$$

This inequality is known as a moving hyperplane, because when the quadrotor moves, the constraint also adaptively changes allowing to approximate the quadrotor and obstacle more precisely and at the same time it prevents the workspace from becoming overly restricted.

The OCP now includes (11) and constraint in (12). This optimization problem is continuous and infinite in time. To address the optimization problem, the OCP is needed to be transformed into the finite nonlinear optimization problem. In this paper, we applied the collocation finite discretization method [10] to discretize the OCP into the nonlinear program (NLP).

III. COLLOCATION FINITE DISCRETIZATION METHOD

The purpose of the discretization method is to convert the optimal control problem (OCP) with infinite dimensions into a nonlinear optimization problem (NLP) with finite dimensions. Firstly, the optimization time interval $[0 \quad t_f]$ is divided into NL time intervals, each treated as an element. The Lagrange polynomials $\ell(t)$ are used to approximate the state variables within each element [10].

$$\begin{aligned} &\Delta t_n = t_{n+1} - t_n;\ n = 1, \dots, \mathrm{NL} \\ &z_n^m(t) = \textstyle\sum_{\mathrm{j=0}}^{\mathrm{NC}} \ell_j(t) \cdot z_{n,j}^m; \\ &m = 1, \dots, \mathrm{ND}; n = 1, \dots, \mathrm{NL} \\ &\ell_j(t) = \textstyle\prod_{\substack{i=0 \\ i \ne j}}^{NC} \frac{t - t_{n,i}}{t_{n,j} - t_{n,i}} \end{aligned} \qquad (13)$$

where $z_n^m(t)$ represents the polynomial approximation of the m-th state variable in the n-th element. $z_{n,j}^m$ is the value of the state variable at the j-th collocation point within the n-th element. $\ell_j(t)$ is the approximation polynomial. Here are some properties of the polynomial $\ell_j(t)$

$$\begin{aligned} &\mathrm{z}_n^m(\mathrm{t_{n,i}}) = \textstyle\sum_{j=0}^{\mathrm{NC}} \ell_j(\mathrm{t_{n,i}}) \cdot \mathrm{z}_{n,j}^m = \mathrm{z}_{n,i}^m \\ &i = 1, \dots, \mathrm{NC} \end{aligned} \qquad (14)$$

Therefore, we have

$$\frac{dz_n^m(t_{n,i})}{dt} = \sum_{j=0}^{NC} \frac{d\ell_j(t_{n,i})}{dt} \cdot z_{n,j}^m \qquad (15)$$

To ensure the continuity of the state variables, the value at the endpoint of one element must be equal to the value at the beginning of the next element. Therefore, we have the following constraint

$$z_{n-1,Nc}^m = z_{n,0}^m$$

Similar to the state variables, algebraic variables are also approximated as follows:

$$y_n^m(t) = \sum_{j=1}^{\mathrm{NC}} \ell_j(t) \cdot y_{n,j}^m$$

$$y_n^m(t_i) = \sum_{j=1}^{\mathrm{NC}} \ell_j(t_i) \cdot y_{n,j}^m = y_{n,j}^m$$

$$m = 1, \dots, \mathrm{ND}; n = 1, \dots, \mathrm{NL}$$

To formulate a general problem that accounts for different time intervals Δt_n, we introduce a common normalized variable $\tau \in [0 \quad 1]$ for each element. This normalized variable is related to the original time Δt_n within each element through the transformation $t = t_n + \Delta t_n \tau$ Consequently, we have the relationship:

$$\frac{dz_n^m}{d\tau} = \Delta t_n \frac{dz_n^m}{dt}$$

Now, by substituting this transformation into Equation (15) and taking the derivative with respect to τ, we obtain the new expression shown as bellows

$$\sum_{\mathrm{j=0}}^{\mathrm{NC}} \frac{\mathrm{d}\ell_j(\tau_{n,i})}{\mathrm{d}\tau} \cdot z_{n,j}^m = \Delta t_{\mathrm{n}} F_m(\mathbf{z}_{\mathrm{n},i}, \mathbf{y}_{\mathrm{n},i}, \mathbf{u}_{\mathrm{n}}, \mathrm{t}_{\mathrm{n},i})$$

In the equation above, $\tau_{n,i} \in [0 \quad 1]$ represents the collocation points i (where $i = 1, \dots, \mathrm{NC}$). In this paper, we choose three collocation points ($NC = 3$) using the Gauss-Legendre method, with $\tau_{n,1} = 0.112702, \tau_{n,2} = 0.500000$, $\tau_{n,3} = 0.887298$. With these collocation points, the approximation error is of the order $O(h^{2\times\mathrm{NC}})$ [10]. Regarding the approximation of state variables, it is described by the following equations

$$\sum_{\mathrm{j=0}}^{\mathrm{NC}} \frac{\mathrm{d}\ell_j(\tau_{n,i})}{\mathrm{d}\tau} \cdot z_{n,j}^m = \Delta t_{\mathrm{n}} F_m(\mathbf{z}_{\mathrm{n},i}, \mathbf{y}_{\mathrm{n},i}, \mathbf{u}_{\mathrm{n}}, \mathrm{t}_{\mathrm{n},i})$$

$$z_{n-1,NC}^m = z_{n,0}^m \tag{16}$$

$$y_n^m(\tau_{n,i}) = \sum_{j=1}^{\mathrm{NC}} \ell_j(\tau_{n,i}) \cdot y_{n,j}^m$$

$$\Delta t_{\mathrm{n}} = \frac{t_f - t_0}{\tau_{n,NC} NL}; \tau_{n,NC} = 0.887298$$

$$i = 1, \dots, \mathrm{NC}; m = 1, \dots, \mathrm{ND}; n = 1, \dots, \mathrm{NL}$$

To simplify, in most problems, the control variable is assumed to be constant within the elements (piecewise constant). For example, during the time interval $[t_n t_{n+1}]$, the value of the control variable U is kept unchanged and treated as a constant. This simplification helps reduce the complexity of the optimal control problem and transforms it into a more manageable finite-dimensional nonlinear optimization problem. With NL elements, the control vector is $\boldsymbol{U} = [U_1, \dots, \boldsymbol{U}_{NL}] \in \Re^{\mathrm{NUx}\,NL}$

IV. NONLINEAR MODEL PREDICTIVE CONTROL ALGORITHM

The aim of the NMPC is to guid the quadrotor to track the reference trajectory while avoiding the collisions with obstacles.

$$E(z,u) = \sum_{n=1}^{NL} \sum_{i=1}^{NC} \left(\left\| z_{n,j}^m - z_r^m \right\|_Q^2 + \|u_n\|_R^2 \right)$$

where $\left\| z_{n,j}^m - z^r \right\|_Q^2 + \|u_n\|_R^2$ is the cost comprising of tracking cost and control cost. The Q and R are defined as the state and control weighting matrices, respectively. z_r^m is the reference path. In addition, the constraints include the equation in (7) and equation in (10). The equation in (9) will determine the lower and upper bound constraints on torques of $T_B, \tau_\phi, \tau_\theta, \tau_\psi$. To this end, the nonlinear optimization problem of the quadrotor can be generally expressed as:

$$\min_{\boldsymbol{z},\boldsymbol{u}} \sum_{n=1}^{NL} \sum_{i=1}^{NC} \left(\left\| z_{n,j}^m - z_r^m \right\|_Q^2 + \|\boldsymbol{u}_n\|_R^2 \right)$$

Subject to

$$\sum_{\mathrm{j=0}}^{\mathrm{NC}} \frac{\mathrm{d}\ell_j(\tau_{n,i})}{\mathrm{d}\tau} \cdot z_{n,j}^m = \Delta t_{\mathrm{n}} F_m(\mathbf{z}_{\mathrm{n},i}, \mathbf{y}_{\mathrm{n},i}, \mathbf{u}_{\mathrm{n}}, \mathrm{t}_{\mathrm{n},i}) \tag{17}$$

$$\boldsymbol{G}(z_{n,j}^m, \mathbf{u}_{\mathrm{n}}) \leq 0$$

$$z_{n-1,NC}^m = z_{n,0}^m$$

$$u_{\min} \leq u_i \leq u_{\max}$$

$$i = 1, \dots, \mathrm{NC}; m = 1, \dots, \mathrm{ND}; n = 1, \dots, \mathrm{NL}$$

The MPC algorithm for UAVs is demonstrated in Table 1. To start the MPC algorithm, the initial positions of these UAVs are given, the NLP is solved where the control and predicted horizon time is set to be equal.

Table 1. The algorithm of the NMPC

initialization*:* Given number of quadrotors (*N*); initial values of state variables $x_{i,0}, y_{i,0}, z_{i,0}$; $i = 1, \dots, N$; positions of obstacles; Control and predicted time horizon.

while $\boldsymbol{k} \leq \boldsymbol{T}$

1. Formulate and solve the NLP in (17) in accordance with control and predicted time horizon to get control quantities $\boldsymbol{u}$
2. Apply the first control vector $\boldsymbol{u}$ to the system
3. $k = k + 1$; update/measure the current positions of the quadrotors
4. Assign the current positions of the quadrotors to $x_{i,0}, y_{i,0}, z_{i,0}$; update the objective function and obstacle avoidance constraints; return to step 1.

End while

V. CASE STUDY

To evaluate the efficacy of the MPC and Obstacle avoidance, the Quad-rotor is controlled to track the orbit, which is expressed as [2].

$$x_r = 5\cos(t/2) - 5$$
$$y_r = 5\sin(t/2)$$
$$z_r = t/2$$

In addition, the quadrotor parameters are given as $I_x = 0.0213; I_y = 0.02217; I_z = 0.0282; b = 4.0687 \times 10^{-7}$

$l = 0.243$; $m = 1.587$; $d = 8.4367 \times 10^{-9}$.

The control system is to designed to ensure that the maximum angular velocity for each motor of 4720rpm. As a result, the following bound constraints are applied for torques [2]:

$$
\begin{aligned}
0 \le T_B \le 36.257 \\
-1.257 \le \tau_\phi \le 1.257 \\
-1.257 \le \tau_\theta \le 1.257 \\
-0.2145 \le \tau_\psi \le 0.2145
\end{aligned}
$$

Moreover, the choice for the weighting matrix is $Q = \text{Diag}\,[1,1,1,0.6,0.6,1,0,0,0,0,0,0]$, while the control input weighting matrix is set as $R = \text{Diag}\,[0.3,0.3,0.3,0.8]$. Additionally, the controller's sampling time is configured at 0.1 seconds, and a prediction horizon spanning 100 samples is employed. The nonlinear optimization problem within the NMPC framework is tackled using the Ipopt solver [9]. Throughout the study, various obstacle avoidance scenarios are simulated, each corresponding to different values of R_{col}.

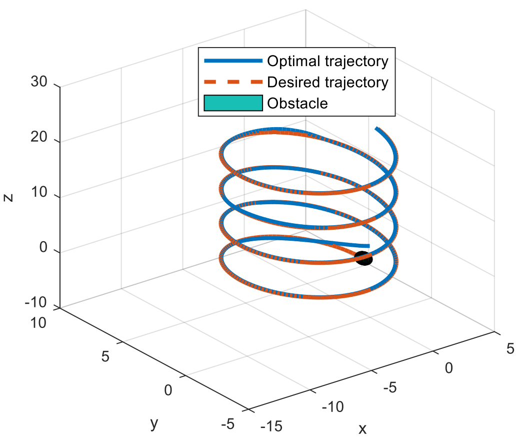

Fig. 2. Optimal quadrotor trajectory

In the initial scenario, the UAV commences from the position [1 1 1], while the obstacle is located at coordinates $x^{ob} = 5\cos(2/2) - 5$; $y^{ob} = 5\sin(2/2)$ and $z^{ob} = 2/2$. The obstacle has a diameter of 0.5, corresponding to $R_{col} = 0.5$. As depicted in Figure 2, the quadrotor adeptly avoids the obstacle while adhering to the desired trajectory.

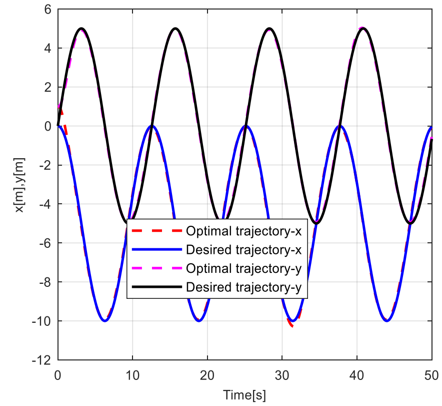

Fig. 3. Optimal trajectory in the x, y axis movement

Furthermore, to demonstrate the effectiveness of our proposed NMPC, the optimal trajectories for each position in the x and y dimensions are visualized alongside the desired trajectories in Figure 3 and Figure 4, respectively. As evident from these figures, the optimal trajectories closely follow the desired paths, further highlighting the robustness of our NMPC approach.

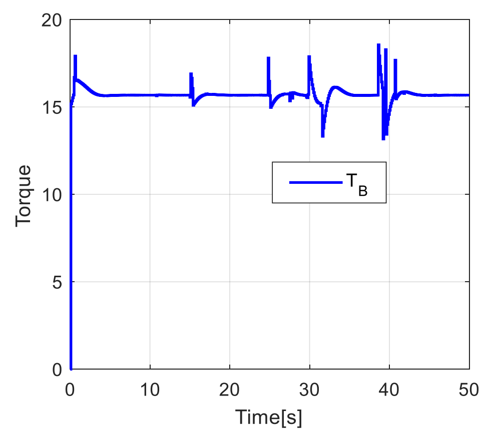

Fig. 4. Optimal thrust torque

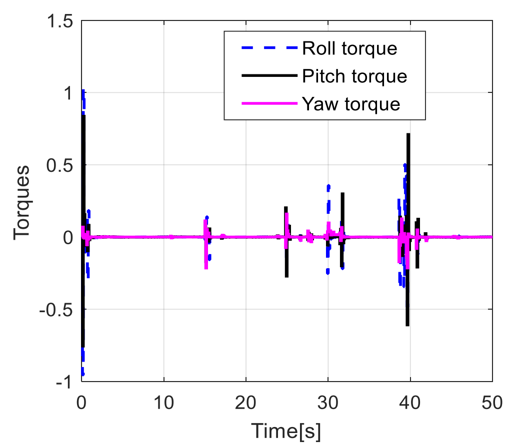

Fig. 5. Optimal roll, pitch, and yaw torques

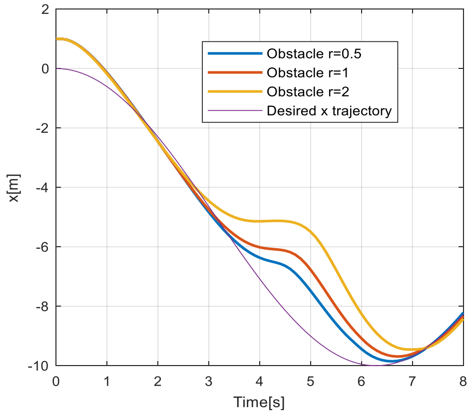

Fig. 6. Obstacle avoidance on the *x* movement

The optimal torques are given in Fig.4 and Fig.5 for thrust torque, the roll, pitch, and yaw torques, respectively. As seen the roll, pitch, and yaw torques are small and in the ranges of bound while the thrust torque is around the value of 15.0.

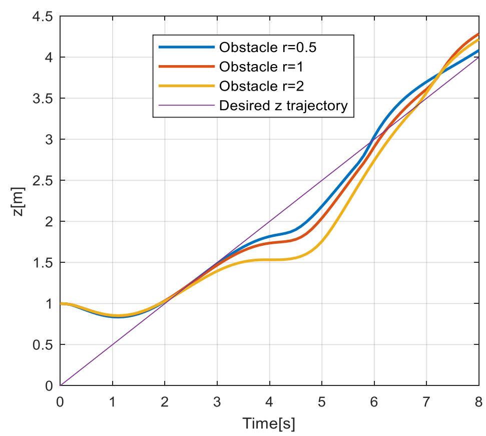

Fig. 7. Obstacle avoidance on the *z* movement

We explore several scenarios with varying values of $R_{col} = 0.5, 1.0, 2.0$. The obstacle is positioned on the desired trajectory at coordinates $x^{ob} = 5\cos(4/2) - 5$; $y^{ob} = 5\sin(4/2)$; $z^{ob} = 4/2$. The effectiveness of the avoidance constraints in the form of moving hyperplanes becomes apparent. Figures 6 and 7 illustrate collision-free trajectories for x and z movements, respectively. These figures demonstrate that the optimal trajectories successfully navigate around the obstacle within the radius of R_{col}. After the avoidance maneuver, the quadrotor resumes tracking the reference trajectory. To further highlight the efficiency of employing hyperplanes for collision-free formulation, the quadrotor's movement in the x-y plane is depicted in Figure 8. This visualization reveals that the optimal trajectories adeptly avoid collision with the obstacle, as indicated by the circles of varying radii.

VI. CONCLUSIONS

We have introduced the utilization of Nonlinear Model Predictive Control (NMPC) to govern the quadrotor's behavior and facilitate obstacle avoidance. This is achieved by employing a moving hyperplane that serves as the basis for formulating linear constraints, essential for obstacle avoidance. The dynamic nature of the hyperplane ensures that it adjusts in accordance with the quadrotor's movements. Consequently, the NMPC framework is capable of regularly updating the collision avoidance constraints while tracking the reference trajectory.

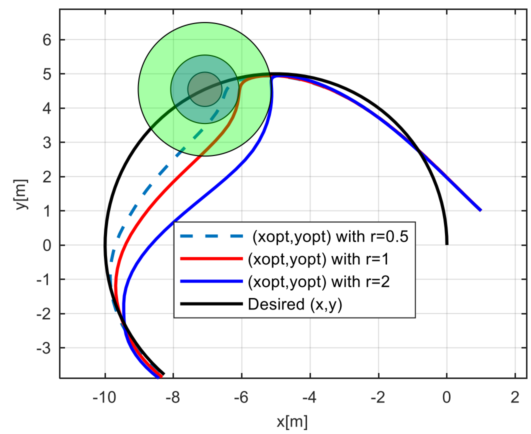

Fig. 8. Trajectories of quadrotor in the *x*-*y* plane

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